Experiment 15: Write a python program to implement Decision Tree

Aim:

To write a Python program to implement a Decision Tree classification algorithm.

Theory:

A Decision Tree is a supervised machine learning algorithm used for classification and prediction. It makes decisions using a tree-like structure consisting of nodes, branches, and leaf nodes.

Algorithm:

1. Start the program.
2. Import the required libraries.
3. Create a dataset with input features and target labels.
4. Split the dataset into training and testing data.
5. Create a Decision Tree classifier.
6. Train the model using the training data.
7. Predict the output for test data.
8. Display the predictions and accuracy.
9. Stop the program.

Python Program:

```
# Alpha-Beta Pruning Algorithm
def alphabeta(depth, node, alpha, beta, is_max, scores):
    if depth == 2:
        return scores[node]
    if is_max:
        best = float('-inf')
        for i in range(2):
            value = alphabeta(
                depth + 1, node * 2 + i,
                alpha, beta, False, scores
            )
```

```
            best = max(best, value)
            alpha = max(alpha, best)
            if alpha >= beta:
                break
        return best
    else:
        best = float('inf')
        for i in range(2):
            value = alphabeta(
                depth + 1, node * 2 + i,
                alpha, beta, True, scores
            )
            best = min(best, value)
            beta = min(beta, best)
            if alpha >= beta:
                break
        return best
scores = [3, 5, 2, 9]
result = alphabeta(
    0, 0, float('-inf'), float('inf'), True, scores
)
print("Best value:", result)
```

exp 15 decession tree.py - C:/Users/yadal/OneDrive/Documents/AI/exp 15 decession tree....

File Edit Format Run Options Window Help

```
# Decision Tree Classification

from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Dataset: [Hours Studied, Attendance]
X = [
    [1, 50],
    [2, 60],
    [3, 65],
    [4, 75],
    [5, 80],
    [6, 90],
    [7, 95],
    [8, 98]
]

# 0 = Fail, 1 = Pass
y = [0, 0, 0, 1, 1, 1, 1, 1]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# Create and train the model
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

# Predict the results
y_pred = model.predict(X_test)

print("Actual Values:", y_test)
print("Predicted Values:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))

# Predict for a new student
result = model.predict([[5, 85]])
```

Sample Output:

```
==== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp  15 decession tree.py ===
Actual Values: [0, 1]
Predicted Values: [0 1]
Accuracy: 1.0
Student Result: Pass
```

Result:

Thus, the Decision Tree classification algorithm was successfully implemented in Python to predict student results based on study hours and attendance